SQL Interview Challenge

Infosys Asked this in 2nd Round

"Given stock prices with missing values (NULL), create a **forward fill** operation: carry the last known price forward until a new price appears."

📊 Stock Prices with Forward Fill

DateKey	StocksName	Price	Derived_Price
2023-01-01	Infosys	1400	1400
2023-01-02	Infosys	NULL	1400
2023-01-03	Infosys	1450	1450
2023-01-04	Infosys	NULL	1450
2023-01-05	Infosys	NULL	1450
2023-01-01	Reliance	2300	2300
2023-01-02	Reliance	NULL	2300

Date Column | Stock Name | Price (Input) | **Derived Price (Output)**

COUNT() OVER Pattern

Universal Forward Fill Solution

1

Count Non-NULL Values

COUNT(Price) increments only for non-NULL prices

→

2

Create Forward Groups

Same count value creates groups for forward fill

→

3

Fill Within Groups

Use FIRST_VALUE or MAX within each group

🎯 **THE KEY INSIGHT**

COUNT() OVER with ORDER BY creates cumulative counts:

1. **COUNT(column) ignores NULLs**: NULL values don't increment the count
2. **ORDER BY matters**: Counts accumulate in chronological order
3. **PARTITION BY resets**: Each stock starts counting from 1
4. **Same count = same group**: All rows with same count belong to same forward fill group
5. **Universal compatibility**: Works in SQL Server 2012+, MySQL 8.0+, PostgreSQL 9.4+, BigQuery, Snowflake

Universal Solution: COUNT() OVER Pattern

SQL Server 2012+, MySQL 8.0+, PostgreSQL

```sql
-- Universal solution using COUNT() OVER
WITH GroupsCTE AS (
    SELECT
        DateKey,
        StocksName,
        Price,
        -- Count non-null prices before current row
        COUNT(Price) OVER (
            PARTITION BY StocksName
            ORDER BY DateKey
        ) AS grp
    FROM
        stock_prices
)
SELECT
    DateKey,
    StocksName,
    Price,
    FIRST_VALUE(Price) OVER (
        PARTITION BY StocksName, grp
        ORDER BY DateKey
    ) AS Derived_Price
FROM
    GroupsCTE
ORDER BY
    StocksName,
    DateKey;
```

Master Forward Fill Patterns

Essential for Time Series Data

The Forward Fill Pattern

Crucial for financial data, sensor data, and time series analysis

```
COUNT() OVER + FIRST_VALUE()
```

✅ Financial Data: Stock prices, currency rates
✅ Sensor Data: IoT device readings
✅ Time Series: Missing data imputation
✅ ETL Pipelines: Data quality cleanup
✅ Analytics: Continuous time series

🔄 **REPOST to help others master this TRICK too !**

💡 Pro Tip: Also learn **backward fill** and **linear interpolation** for complete time series handling!